

Assignment-3

Q.1) MATLAB Code for Simulation of Temperature Profiles:-

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clear; clc; close all;

%% Parameters & Assumptions
L = 5; % Length of heat exchanger (m)
x_span = [0 L]; % Spatial domain
U_P = 500; % Product of overall heat transfer coeff
and perimeter (W/mK)
% Heat capacity rates (mass flow * specific heat) in W/K
Ch = 1000; % Hot fluid
Cc = 800; % Cold fluid
% Inlet Temperatures (K)
Th_in = 373.15; % 100 C
Tc_in = 293.15; % 20 C

%% Parallel-Flow Simulation
% y(1) = Th, y(2) = Tc
% Equations: dTh/dx = -U*P/Ch*(Th-Tc); dTc/dx = U*P/Cc*(Th-Tc)
parallel_ode = @(x, y) [-(U_P/Ch)*(y(1) - y(2));
                        (U_P/Cc)*(y(1) - y(2))];
% Solve using ODE45 (Initial conditions: both inlets at x=0)
[x_p, temp_p] = ode45(parallel_ode, x_span, [Th_in; Tc_in]);

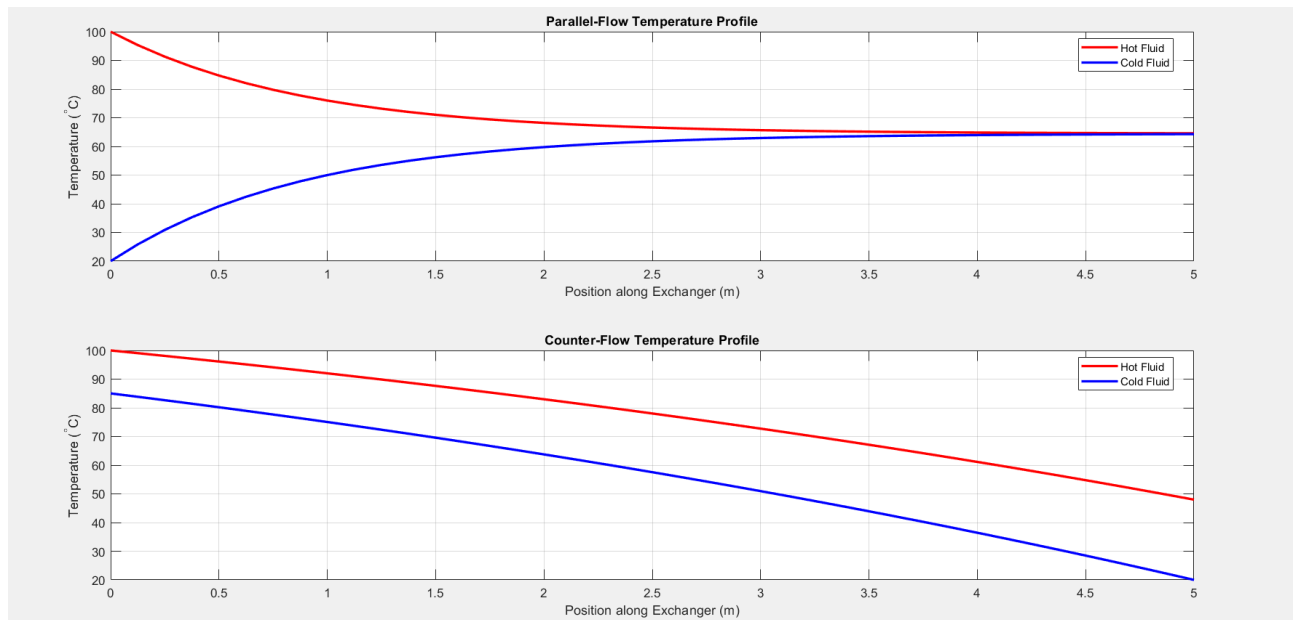
%% Counter-Flow Simulation
% For counter-flow, Tc is known at x=L. We use bvp4c (Boundary
Value Problem solver)
% Equations: dTh/dx = -U*P/Ch*(Th-Tc); dTc/dx =
-U*P/Cc*(Th-Tc)
counter_ode = @(x, y) [-(U_P/Ch)*(y(1) - y(2));
                        -(U_P/Cc)*(y(1) - y(2))];
% Boundary conditions: Th at x=0 (ya) and Tc at x=L (yb)
bcs = @(ya, yb) [ya(1) - Th_in; % Th(0) = 100C
                 yb(2) - Tc_in]; % Tc(L) = 20C
% Initial guess for the solver
solinit = bvpinit(linspace(0, L, 10), [Th_in, Tc_in]);
sol_c = bvp4c(counter_ode, bcs, solinit);
% Evaluate solution for plotting
x_c = linspace(0, L, 100);
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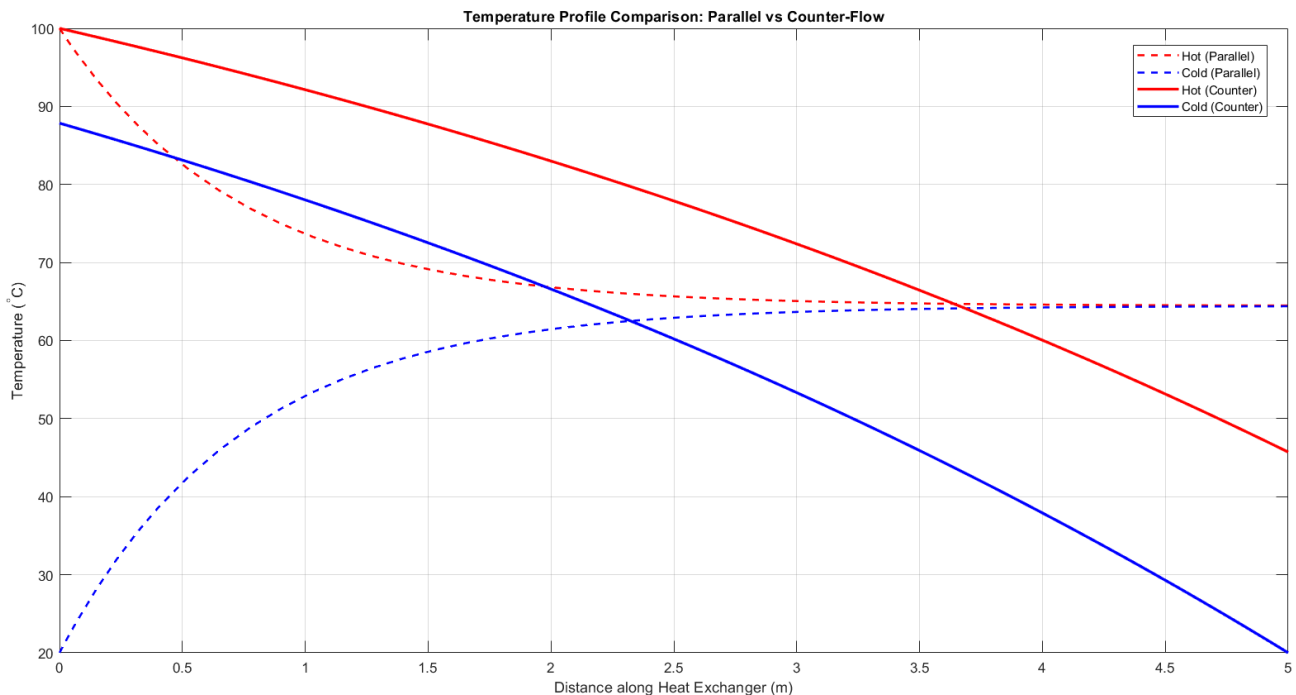
temp_c = deval(sol_c, x_c);

%% Plotting Results
figure('Name', 'Heat Exchanger Temperature Profiles',
'NumberTitle', 'off');
% Parallel Flow Plot
subplot(2,1,1);
plot(x_p, temp_p(:,1) - 273.15, 'r', 'LineWidth', 2); hold on;
plot(x_p, temp_p(:,2) - 273.15, 'b', 'LineWidth', 2);
title('Parallel-Flow Temperature Profile');
xlabel('Position along Exchanger (m)');
ylabel('Temperature (^{\circ}C)');
legend('Hot Fluid', 'Cold Fluid', 'Location', 'best');
grid on;
% Counter Flow Plot
subplot(2,1,2);
plot(x_c, temp_c(1,:) - 273.15, 'r', 'LineWidth', 2); hold on;
plot(x_c, temp_c(2,:) - 273.15, 'b', 'LineWidth', 2);
title('Counter-Flow Temperature Profile');
xlabel('Position along Exchanger (m)');
ylabel('Temperature (^{\circ}C)');
legend('Hot Fluid', 'Cold Fluid', 'Location', 'best');
grid on;

```



Q.2) Plot of both profiles on the same graph:-



- a) **Higher Outlet Cold-Fluid Temperature:** The **Counter-flow** configuration yields a higher outlet cold-fluid temperature. In parallel flow, the cold fluid's exit temperature is limited by the hot fluid's exit temperature. In counter-flow, the cold fluid exits near where the hot fluid enters, allowing it to reach a much higher final temperature.
- b) **Temperature Driving Force:** The driving force ($\Delta T = T_h - T_c$) is better maintained in **Counter-flow** because the temperatures of the two fluids change in a way that keeps the difference between them more uniform across the entire length. In parallel flow, ΔT is very large at the inlet but drops off rapidly, making the latter half of the heat exchanger much less effective.

Q.3) Special Case Analysis:-

```
clear; clc; close all;
%% Parameters & Assumptions
L = 5; % Length (m)
x_span = [0 L]; % Spatial domain
U_P = 600; % Heat transfer coefficient * perimeter
Ch = 1000; % Hot fluid capacity rate (W/K)
Cc = 800; % Cold fluid capacity rate (W/K)
Th_in = 100; % Inlet Hot (C)
Tc_in = 20; % Inlet Cold C

%% Simulation Logic (Parallel and Counter)
% Parallel Flow ODE
ode_p = @(x, T) [-(U_P/Ch)*(T(1)-T(2)); (U_P/Cc)*(T(1)-T(2))];
[x_p, temp_p] = ode45(ode_p, x_span, [Th_in; Tc_in]);
% Counter Flow BVP (Boundary Value Problem)
ode_c = @(x, T) [-(U_P/Ch)*(T(1)-T(2));
-(U_P/Cc)*(T(1)-T(2))];
bc_c = @(Ta, Tb) [Ta(1) - Th_in; Tb(2) - Tc_in]; % Hot at
x=0, Cold at x=L
solinit = bvpinit(linspace(0, L, 5), [Th_in, Tc_in]);
sol_c = bvp4c(ode_c, bc_c, solinit);
x_c = linspace(0, L, 100);
temp_c = deval(sol_c, x_c);

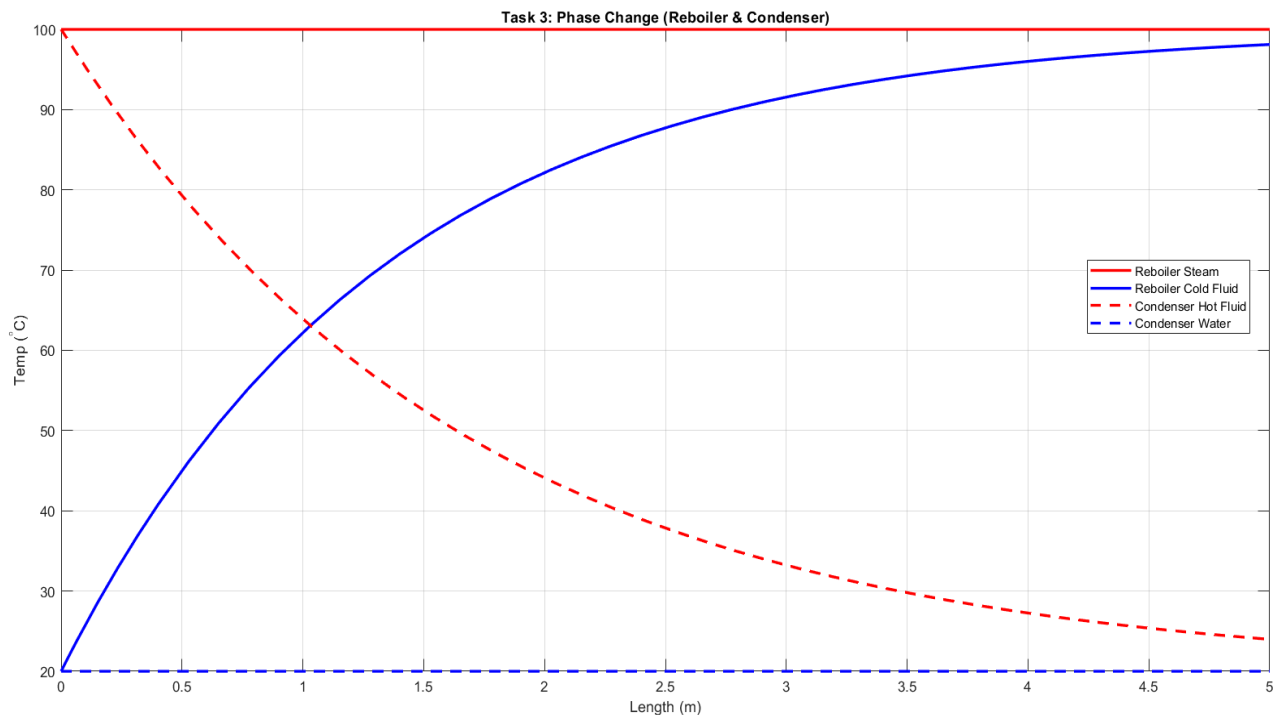
%% Special Case Analysis (Reboiler & Condenser)
% Reboiler: Hot fluid is constant (Phase change)
ode_reboil = @(x, Tc) (U_P/Cc)*(Th_in - Tc);
[x_reb, T_reb] = ode45(ode_reboil, x_span, Tc_in);
% Condenser: Cold fluid is constant (Phase change)
ode_cond = @(x, Th) -(U_P/Ch)*(Th - Tc_in);
[x_con, T_con] = ode45(ode_cond, x_span, Th_in);

%% Plotting Results
figure('Color', 'w', 'Position', [100 100 1000 700]);
% Special Cases
plot(x_reb, ones(size(x_reb))*Th_in, 'r-', x_reb, T_reb, 'b-',
'LineWidth', 2); hold on;
plot(x_con, T_con, 'r--', x_con, ones(size(x_con))*Tc_in,
'b--', 'LineWidth', 2);
title('Task 3: Phase Change (Reboiler & Condenser)');
xlabel('Length (m)'); ylabel('Temp (^{\circ}C)');
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legend('Reboiler Steam', 'Reboiler Cold Fluid', 'Condenser Hot Fluid', 'Condenser Water', 'Location', 'best');
grid on;

```



Q.4)

a) Phase-Change Profiles: Phase-change cases do not follow a standard dual-exponential profile because one fluid remains at a constant temperature (the saturation temperature) throughout the exchanger. Heat transfer only changes the phase (latent heat) of that fluid rather than its temperature.

b) Simplified LMTD: Log Mean Temperature Difference (LMTD) becomes simpler for reboilers and condensers because one of the temperature differences (ΔT_1 or ΔT_2) is based on a constant value. Mathematically, the flow arrangement (parallel vs. counter) becomes irrelevant to the calculation since the

constant-temperature fluid provides the same profile regardless of direction.